THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · HARD LEVEL CHAPTER TEST SET 3

Chapter 1: **Coordinates** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If the distance between $(x, -4)$ and $(3, 11)$ is 17 and $x > 3$, then $x =$ 1

(a) 11 **(b)** 10 **(c)** 13 **(d)** 12

2. The fourth vertex of the parallelogram with consecutive vertices $(0, -1)$, $(3, -2)$, $(5, 2)$ (in order) is: 1

(a) $(3, 2)$ **(b)** $(-2, 3)$ **(c)** $(2, 3)$ **(d)** $(2, -3)$

3. A point $P(x, 0)$ on the x-axis is 29 units from $(0, 21)$ with $x > 0$. The value of x is: 1

(a) 19 **(b)** 21 **(c)** 22 **(d)** 20

4. The points $(-4, 5)$, $(0, 5)$ and $(7, 5)$ lie on a: 1

(a) vertical line **(b)** line through the origin

(c) line in Quadrant III only **(d)** horizontal line

5. The area of the triangle with vertices $(0, 0)$, $(24, 0)$ and $(0, 7)$ is: 1

(a) 84 **(b)** 86 **(c)** 83 **(d)** 85

6. The distance of the circumcentre of the right triangle with vertices $(0, 0)$, $(14, 0)$ and $(0, 48)$ from the origin is: 1

(a) 25 **(b)** 27 **(c)** 24 **(d)** 26

7. A circle has diameter with endpoints $(2, 2)$ and $(34, 62)$. Its radius is: 1

(a) 36 **(b)** 35 **(c)** 34 **(d)** 33

8. The point on the y-axis equidistant from $(3, -4)$ and $(-3, 2)$ is: 1

(a) $(1, -1)$ **(b)** $(-1, 0)$ **(c)** $(0, 1)$ **(d)** $(0, -1)$

9. **Assertion (A):** The reflection of the point $(3, -5)$ in the x-axis is $(3, 5)$. 1

Reason (R): The reflection of a point in the x-axis keeps its abscissa and changes the sign of its ordinate.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** The distance between the points $(1, 2)$ and $(4, 6)$ is 5 units. 1
Reason (R): The distance between two points on the plane is never negative.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the distance between the points $(-3, -6)$ and $(6, 6)$. 2

12. The vertices of a right triangle are $O(0, 0)$, $A(14, 0)$ and $B(0, 48)$. Find the circumcentre and the circumradius. 2

13. The end points of a diameter of a circle are $A(3, -3)$ and $B(27, 29)$. Find the centre and the radius. 2

14. Find the distance between the points $(8, 8)$ and $(-8, -4)$. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. Find the points on the x-axis whose distance from $A(-5, 24)$ is 30 units. 3

16. Three vertices of a rectangle $ABCD$ are $A(0, 4)$, $B(24, 36)$ and $D(-24, 22)$. Find C and the diagonal AC. 3

17. Find the length of the median AD of the triangle with $A(0, 0)$, $B(54, 0)$, $C(0, 72)$, where D is the midpoint of BC. 3

OR

Find the points on the x-axis whose distance from $A(3, 30)$ is 34 units. 3

18. Three vertices of a rectangle $ABCD$ are $A(-1, -1)$, $B(35, 47)$ and $D(-21, 14)$. Find C and the diagonal AC. 3

OR

Find the length of the median AD of the triangle with $A(0, 0)$, $B(60, 0)$, $C(0, 80)$, where D is the midpoint of BC. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A rectangular garden is drawn on a coordinate plane (1 unit = 1 m) with corners P $(0, 0)$, Q$(8, 0)$, R$(8, 6)$ and S$(0, 6)$. A sprinkler is to be fixed at the midpoint of the diagonal PR. 5
(i) Find the length of PQ.
(ii) Find the coordinates of the midpoint of PR.
(iii) Find the length of the diagonal PR.
(iv) Show that the diagonal QS has the same length and the same midpoint as PR.

20. Show that $A(0, 5)$, $B(9, 17)$, $C(-3, 26)$, $D(-12, 14)$ are the vertices of a square. Find its side, diagonal squared and area. 5

OR

The point $P(x, 0)$ is equidistant from $A(-15, 5)$ and $B(9, -5)$. Find P and the distance PA. 5

END OF QUESTION PAPER Total: 40 Marks